

Research Article

The longitudinal interplay of print exposure and auditory brain responses in the prediction of reading skills

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ABSTRACT

Auditory brain responses' longitudinal interplay with print exposure and their predictive effects on reading skills were examined in a sample followed from age 4 to 10 years. Print exposure was represented by parent–child shared reading and children's independent print exposure, whereas speech and non-speech stimuli were used to evoke brain responses (N250). We also examined the roles of familial risk (parental dyslexia) and parental education in the path models. The results suggested significant links between print exposure and brain responses to speech. A reciprocal association existed between independent print exposure and brain responses to speech, whereas unidirectional path was found from non-speech brain responses to independent print exposure. Independent print exposure in turn predicted subsequent reading fluency. Increased Parent-child shared reading at the onset of formal literacy schooling was associated with more negative brain responses to speech stimuli. Family risk for dyslexia predicted poorer reading accuracy and fluency, and the effect did not run via print exposure or brain responses. These results suggest significant longitudinal associations between brain responses to auditory stimuli, print exposure, and reading outcomes implicating evidence for the brain-environment association. The precise underlying mechanisms require further investigation.

1. Introduction

Children are subject to manifold influences during their development towards becoming proficient readers. Early print exposure provided prior to formal literacy instruction are associated with reading development, together with parental reading skills and socio-economic-status of the parents (e.g., Hart et al., 2021; Little et al., 2021; Lohvansuu et al., 2021). Print exposure provided at home in childhood can appear in different forms, such as parent–child shared reading or children's independent print exposure. From the point of view of influence on brain

development for reading, these two forms of print exposure have conceptually different attributes. Shared book reading is an act of parent–child interaction that includes a visual–auditory integration of the parents' voice into the print or picture material and frequently discussions between child and parent. On the contrary, children's independent interactions with print materials are a typically silent activity in which they examine pictures and print and are also possibly recalling the storylines that had been previously introduced to them. Both print exposure forms are shown to be important for the development of reading (Church et al., 2023; Mol & Bus, 2011; Scarborough & Dobrich,

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1994; Sun et al., 2024). However, print exposure may not have the same impact on all children as also the state of the brain can affect the environmental effect. The brain development of typical readers has revealed to be characterized by the emergence of neural circuits for reading (Ozernov-Palchik & Gaab, 2016), structural maturation and increased structural connectivity of the left language areas (Chyl et al., 2021), with evidence found in both print and speech processing alongside evidence of coactivation of these areas (Chyl et al., 2018). Brain maturation and development of different skills to certain degrees of specialization, e.g. the development of the visual word form area (McCandliss et al., 2003), are needed for children to gain expertise in reading. The shift from random to more organised functional networks is considered part of brain maturation (Boersma et al., 2011). The processes between print exposure and brain development are, however, yet to be understood better as environmental and brain-related factors on reading development have mostly been studied separately. Specifically, little is known about whether sound related brain responses, which have been seen as critical indicators of reading development and dyslexia (Volkmer & Schulte-Körne, 2018), are influenced by or associated with print exposure across time and reading development. Vice versa, the role of the brain responses to speech and non-speech sounds in engagement in print exposure is yet mostly unclear. Thus, the novelty of this study lays in its triangulation of environmental, neural and skill factors within a longitudinal modelling approach with autoregressive controls, conceptualizing the developmental processes as an interplay between factors. The longitudinal nature of this study allows an evaluation of the connections between print exposure (shared parent-child reading and child's independent reading frequency), brain responses to auditory stimuli (speech and non-speech) during a critical developmental period for literacy development between ages 4 and 10. Furthermore, we examine the role of parental education and dyslexia because both parental education and family risk for dyslexia (e.g., Khanolainen et al., 2020; Lohvansuu et al., 2021) are found to predict children's reading skills.

1.1. Print exposure and reading development

Engagement with language and print material is needed for the developing reader to enhance their literacy skill. Empirical studies have reported robust evidence that, at the behavioral level, the association between print exposure and reading skills does exist (Mol & Bus, 2011; Scarborough & Dobrich, 1994). Parent-child shared book reading, a central part of the home literacy environment model (Sénéchal & LeFevre, 2002, 2014), has been repeatedly shown to contribute to children's language and reading development across different linguistic contexts (Mol et al., 2008; Noble et al., 2019; Savva et al., 2022; Torppa et al., 2020, 2022; van Bergen et al., 2021). Although some studies have reported mixed findings on the direct effect of HLE activities on reading skills (Georgiou et al., 2021), systematic reviews indicate that the association is quite robust (e.g. Mol & Bus, 2011; Nag et al., 2024). One beneficial aspect of shared reading is suggested to be the possibility for high-quality linguistic inputs that book language offers alongside parent-child interactions (Mol & Bus, 2011; Rowe & Snow, 2020). This linguistic aspect is accompanied by the developing readers' attention towards pictures and print, which has been shown to increasingly shift towards print with developing reading skills (Roy-Charland et al., 2007).

Apart from parent-child shared reading activities, children also engage independently with print materials. This form of print exposure with age-appropriate texts has been identified as a rich source for the increase of print awareness, with its increased encounter to words, sentence structure and unknown vocabulary, providing a rich environment for the developing reader (Mol & Bus, 2011). A type of upward spiral has been identified for proficient readers, where the process of experiencing pleasure during the reading exercise, increases the likelihood of them choosing to read more often and therefore having a positive effect on their improvement in language skills (Mol & Bus, 2011), it

is further said to increase the quality of word form representations and thus increase the ability of word recognition. Previously it was shown in a study from the same sample as the present (van Bergen et al., 2021) that children's independent print exposure and reading skills were reciprocally associated across the ages of 5–15 years. This finding is compatible with the view of children becoming more active in shaping their own environment by seeking out opportunities that enrich their environment with increasing skill (Mol & Bus, 2011).

1.2. ERP and reading development

Several earlier studies have used electroencephalography-based event-related potentials (ERPs) to examine group differences in children with different reading skills and found associations with reading skills (Ozernov-Palchik & Gaab, 2016; Volkmer & Schulte-Körne, 2018). Most of the longitudinal studies have examined auditory ERPs as they are easily measured before reading instruction begins, even in infancy. Auditory speech sounds have been thought to link with phonological processing, phonological processing being an important factor for reading and reading development. On the contrary, auditory non-speech can capture sound processing that is less tightly linked to phonological processing, explaining independent variance. The line of non-speech dyslexia research is based on the idea that speech relies on identifying basic key features in sound signal which in turn could lead to the assumption of more general auditory processing being involved. In our Finnish samples, poorer reading scores have been linked to a basic auditory ERP response, the N250 (Azaiez et al., 2022; Hämäläinen et al., 2013; Lohvansuu et al., 2014). Group differences have also been found between children with and without dyslexia in the N250 responses which are thought to reflect functional and possibly structural differences which also relate to children's reading skills (Lohvansuu et al., 2014). Additionally, the N250 responses have been positively correlated with fixation-related visual N170 responses in children during sentence reading (Azaiez et al., 2022), further supporting the idea that the N250 relates to the proficiency in reading for the developing reader. The link between the auditory sensory processing and the development of reading skill and family risk for dyslexia makes the neural processing of speech sounds an ideal candidate to be investigated alongside the environmental factor of print exposure.

1.3. Print exposure and brain measures

As illustrated in the previous sections, print exposure and ERP responses to auditory stimuli have been linked to the reading skill development of children. Despite these findings, studies that include brain and skill measures alongside environmental effects are few.

Indications on the association between home learning environment and ERP responses to speech sounds was found in a sample of three-year-olds when separating the group in high and low stimulating environments and the groups showing distinguishable speech processing in their ERP responses (Molfese & Molfese, 2002). Other connections of EEG features and environmental factors have been proposed in the literature: Brito and colleagues (2020), for example, examined the link between EEG beta power in infants and adult word count in their environment, and found that children who heard more adult words tended to have reduced beta power in high-chaos homes.

There is evidence for the association between brain measures and a more distal factor of children's environments, the socioeconomic status (SES) (Bruto & Noble, 2014; Hackman & Farah, 2009; Rakesh et al., 2021; Rakesh & Whittle, 2021). The effect of SES on the individuals' development has been proposed to run through the proximal environment of the child, through factors such as HLE (Romeo et al., 2022). The few studies incorporated brain measures and HLE activities, such as shared reading, have suggested that children's engagement in parent-child shared reading is associated with greater functional connectivity of language areas and right hemispheric brain areas (Hutton et al.,

2017a, 2017b), strengthening the concept of environmental factor being associated with neural development.

Also the frequency of shared reading and literacy teaching activities has been linked to different levels of activation in the brain's reading network (Girard et al., 2021; Hutton et al., 2015, 2020; Powers et al., 2016). In a longitudinal examination of shared reading and white matter organisation, the extent of the left lateralisation of the superior longitudinal fasciculus (SLF) was found to mediate the relationship between shared reading in kindergarten and reading skills at the Grade 2 level (Davison et al., 2023). SLF has also been associated with speech perception and reading abilities (Davison et al., 2023).

Furthermore, children's brain responses to shared reading have been investigated in laboratory settings wherein mothers were asked to read to their children (Hasegawa et al., 2021) or children were asked to listen to story recordings (Hutton et al., 2015). Between unknown experimenters and the familiar voices of their mothers, the mothers' reading elicited enhanced functional brain connectivity among the children and triggered the higher attention paid by the latter (Hasegawa et al., 2021). Aside from evidence indicating the specific effects of familiar voices on children's brain activation in shared reading, another study found that the activation level of the left-sided semantic language processing hub (parietal-temporal-occipital association area) during the standardised storytelling of a female voice was positively correlated with higher print exposure at home (Hutton et al., 2015). Already in infancy, the connectivity levels of brain structures linking regions of speech production and language understanding (left arcuate fasciculus) were associated with parent-child reading-related activities (Turesky et al., 2022). These findings indicate the possibility that home reading environments have an impact on the development of brain areas related to language processing and reading.

1.4. Present study

The present study investigates a model, that includes a proxy for children's genetic dispositions (family risk for dyslexia), incorporates children's print exposure and recorded brain responses to speech-related stimuli over time and examines their association with later reading skills. In combination with the intergenerational transmission of dyslexia (i.e. family risk) (van Bergen et al., 2014) and factors accounted for in the cumulative risk and resilience model (e.g. environmental influences) (Catts & Petscher, 2022). To add to the knowledge from the previous studies the hypothesised model enables us to examine the

longitudinal associations between print exposure and brain responses to speech and non-speech stimuli (see Fig. 1 for the hypothesised model).

For the longitudinal interplay of print exposure and brain activation to speech, we posit that earlier print exposure (i.e. shared reading and children's independent print exposure) can predict the brain's responses to speech and that brain activation subsequently predicts later print exposure, which is being evaluated in the link to the outcome of print exposure. Another important aspect of the model is the examination of change of relationship of print exposure and ERP over time, which is being accounted for by including two time points of ERP and print exposure and by controlling for the autoregressive effects. This feature allows us to tap developmental change and the effect of the other measures on change. The model also allows examination of mediated effect from HLE to brain responses. We hypothesise that the predictive effect of print exposure on later reading skill outcomes occurs partly via the specialisation that is reflected in the brain responses to speech stimuli. Further, the direct and mediated effects of familial risk for dyslexia are examined. Based on previous findings, we posit that family risk for dyslexia predicts ERP (Hämäläinen et al., 2013) and reading scores, but has no direct influence on print exposure (Torppa et al., 2007, 2022). In contrast, we hypothesise that parental education predicts print exposure (Lyytinen et al., 1998; Myrberg & Rosén, 2009; Sénéchal & LeFevre, 2002). In addition to the model incorporating brain responses to speech, we evaluate the non-speech model to examine whether the association between print exposure and brain responses in predicting reading is present only for speech or whether it is also more generally seen for auditory processing. Thus, the need for two models (ERP speech and ERP non-speech) emerged due to the theoretical differentiation between speech and non-speech stimuli in dyslexia brain research (Schulte-Körne & Bruder, 2010). For the environmental aspects, we focus on two types of print exposure (i.e. parent-child shared reading and children's independent print exposure) due to the potentially different mechanisms that they introduce. Shared reading is a form of print exposure that carries linguistic and orthographic auditory input as well as the social aspect that allows e.g. discussions around the print material. In comparison, independent print exposure includes solitary engagement with print/pictures, which does not as likely as in shared reading include the auditory aspect, and particularly among the younger children in this sample who are not yet readers (age 4–5) does not include actual reading but rather looking at pictures and print and possibly inventing stories or recalling the story heard before. This differentiation is a conceptual differentiation that is not directly verifiable

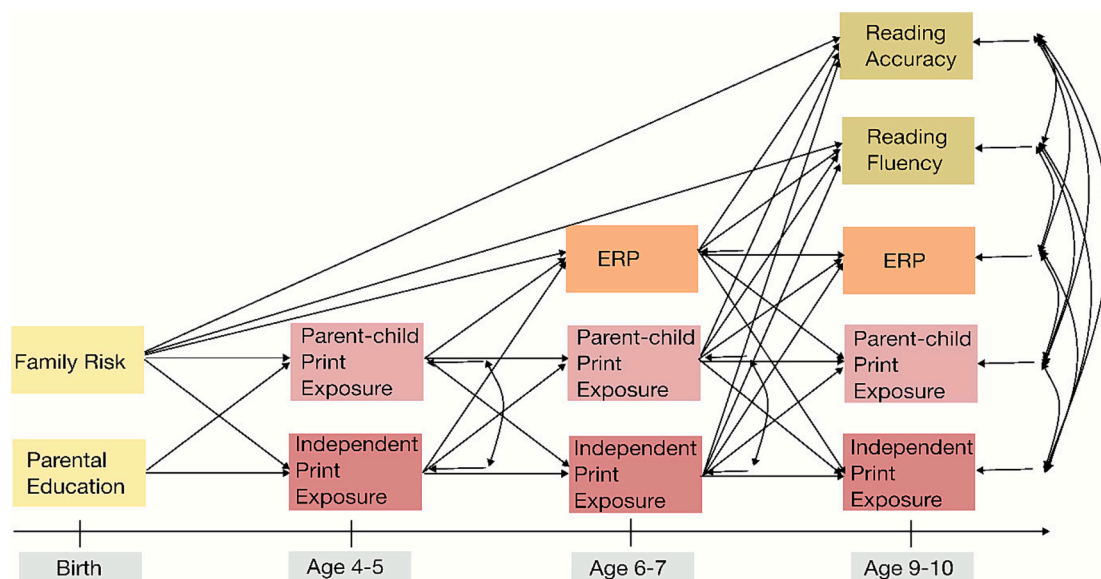


Fig. 1. Hypothesised model for event-related potential (ERP), print exposure during parent-child shared reading and children's independent print exposure.

in the presented print exposure data obtained from parental questionnaires.

2. Methods

2.1. Participants

Data were collected within the Jyväskylä longitudinal study of dyslexia (JLD) (Lohvansuu et al., 2021), where participants' development was followed from infancy to adulthood. The sample included families with and without family risk for dyslexia (based on tests and self-reports of the parents). In the present analysis, we focused on the period from ages 4–10 to examine children's development from pre-literacy to literacy. All children included in this data analysis entered school in August of the year they turned 7 years old, as children do in the Finnish educational system. Prior to Grade 1, they received no formal reading instruction. The native language of all participants was Finnish. In an earlier investigation, identification of the participating children with dyslexia was performed at Grade Two (Eklund et al., 2015), providing grounds for a participant pool with diverse reading skills. A positive ethical statement was given by the ethical board of the Central Finland Hospital District. All parents and later, their children, gave informed consent for participation. The participation of children and their parents, over time, resulted in varying amounts of data per variable entered in the examined model: reading skills outcome at age 9 was at the higher edge of participation ($N = 191$), whereas ERP variables for speech differed between the ages 6.5 ($N = 64$) and 9 ($N = 138$). All descriptive statistics and Pearson's correlations of the model variables are given in Table 1. Skewness values of the variables are based on the type 2 method (Joanes & Gill, 1998).

2.2. Family risk for dyslexia

Family risk for dyslexia was determined based on parental dyslexia: if the father or mother had dyslexia based on their cognitive test scores (prenatal to the birth of the to-be-participating child), reported having a history of reading problems and had a relative who also had reading difficulties, the status of family risk was given for the child. The cognitive test scores utilized for parental dyslexia identification included measures of phonological decoding, orthographic processes, oral text reading and spelling (Leinonen et al. 2001). The parental recruitment phase and the exact criteria have been described in depth in previous

publications (Eklund et al., 2015; Leinonen et al., 2001).

2.3. Parental education

Parental education is a composite score of both parents' educational levels at the parental recruitment stage. The participants' parents had to indicate their level of education on a six-point scale ranging from comprehensive school only to a higher university degree.

2.4. Measures of print exposure

Print exposure was measured through parental reports in two distinct categories: independent print exposure and shared book reading with parents. The latter reflects the frequency and duration of shared reading with parents, whereas the former reflects the frequency and duration at which children spent interacting with books/magazines on their own. Both were based on parental reports (Fig. 2), presenting a coarse measure of print exposure. Print exposure with parents was examined at ages 4–7, and age 9. At ages 4 and 5, print exposure with parents was reported via five items, and at ages 6, 7 and 9 with four items per time point. Each item of print exposure was examined with five-point Likert scales. Three mean composite scores were calculated for parent–child shared reading: one for ages 4–5 (Cronbach's $\alpha = 0.86$), another one for ages 6–7 (Cronbach's $\alpha = 0.77$), and one for age 9 (Cronbach's $\alpha = 0.75$). The independent print exposure was consistently measured at ages 4–9 with three items using a five-point Likert scale. Three mean composite scores were calculated for independent print exposure: one for ages 4–5, another one for ages 6–7, and one for age 9. The Cronbach's α values ranged between 0.70 and 0.82 within these time-points. Earlier examination of the same cohort focused on home literacy environments effect on reading development (Torppa et al., 2022).

2.5. Reading accuracy and fluency measures

Reading accuracy and fluency levels at age 9 were included as measures of reading skills as outcomes. Each of these items was a composite score of three reading tasks: oral text reading, oral pseudoword text reading and word list reading. In each task, the child was instructed to orally read the word/text as fast and accurately as they could. Oral text reading was examined using a text with 189 words presented in 18 sentences. Oral pseudoword text reading consisted of 38

Table 1
Correlation table.

Variable	Valid N	Skew	M	SD	1	2	3	4	5	6	7	8	9	10	11
1. Independent print exposure 4–5	183	0.00	2.91	0.76											
2. Independent print exposure 6–7	168	-0.13	2.45	0.60	.66**										
3. Independent print exposure 9–10	177	-0.44	2.74	0.71	.53**	.65**									
4. Parent-child print exposure 4–5	174	-0.21	3.12	0.64	.50**	.28**	.34**								
5. Parent-child print exposure 6–7	155	0.05	2.35	0.55	.42**	.30**	.26**	.75**							
6. Parent-child print exposure 9–10	165	0.41	1.73	0.61	.25**	.12	.04	.51**	.65**						
7. ERP speech 6.5	64	-0.00	-0.27	0.89	-.28*	-.25	.37**	-.14	-.06	.05					
8. ERP non-speech 6.5	63	-0.14	-0.29	0.99	-.11	-.10	-.29*	.05	.18	.10	.67**				
9. ERP speech 9–10	138	-0.78	-3.27	2.29	-.03	.00	-.03	.23*	.26**	.11	.39**	.45**			
10. Reading fluency 9–10 (z.)	191	0.39	0.00	1.0	.07	.24**	.41**	.04	.01	-.26**	-.21	-.16	-.07		
11. Reading accuracy 9–10 (z.)	191	-1.46	0.00	1.0	-.05	-.01	-.14	.02	-.05	-.22**	-.25*	-.37**	-.15	.51**	
12. Parental education	198	0.66	4.03	1.10	.05	-.01	.09	.06	.14	.02	-.04	.15	-.04	.09	.07

Note. Skew stands for skewness, M and SD represent mean and standard deviation. The indication z stands for z-scored values. Valid N is indicating the number of observations of each of the variables. Here * indicates $p < 0.05$., and ** indicates $p < 0.01$.

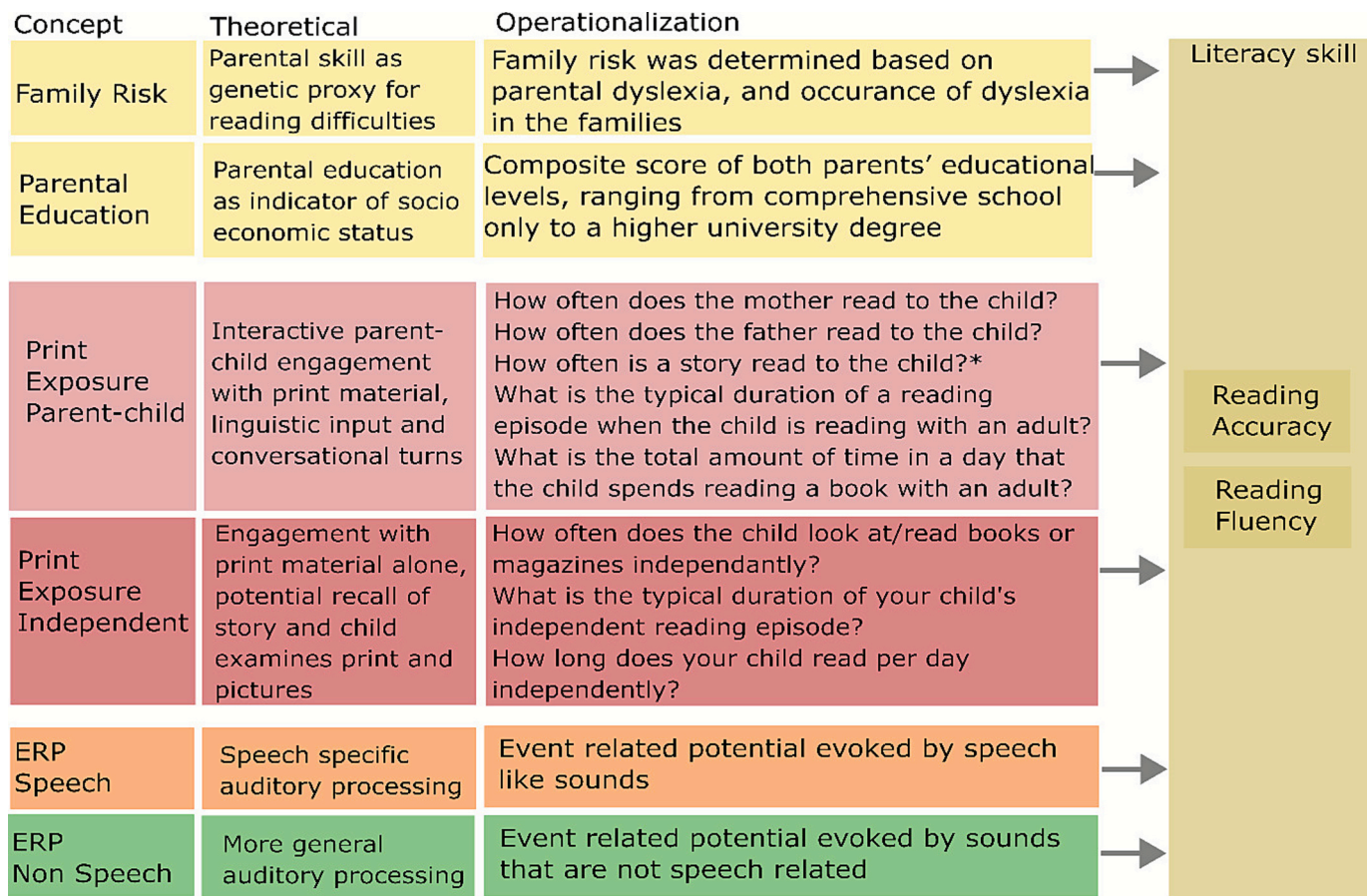


Fig. 2. Schematic overview of the concepts of interest in this study, namely concepts given by parents, print exposure and ERP brain responses. The concepts of print environment are split into print exposure as parent-child activity and independent child activity, questionnaire items are given, * indicates the item that was not available for inclusion in the age 6–9 timepoints.

pseudowords. In both text reading tasks, the time to complete the text was taken. For the word list reading, a 105-item list was utilized, and participants were given 2 min to read as many items aloud as they could. The item length increased from 3 to 22 letter words. Reading accuracy score was the mean score of the percentage of correctly read words per task. The reading fluency score was the mean score of read letters per second per task. The reading skill variables used in the presented model were transformed into z scores ($M = 0$, $SD = 1$). The Cronbach's alpha for reading accuracy composite was 0.83 and 0.89 for the reading fluency composite, more details on the reading skills examination are given in an earlier publication regarding this longitudinal investigation (Eklund et al., 2015). The motivation for including both, pseudoword and word text reading anchored in the constitution of Finnish language, with earlier research suggesting their high correlation and internal consistency.

2.6. Brain data measures

The brain measure incorporated in the current study was the auditory ERP response N250, a negative response that reflected auditory sensory processes and demonstrate maturational effects (Čeponienė et al., 2002). It is evoked by the standard and the deviant stimulus, in contrast to the pre-attentive change detection response referred to as mismatch negativity (MMN) (Näätänen et al., 2007; Ozernov-Palchik & Gaab, 2016; Schulte-Körne & Bruder, 2010), which reflects the brain's ability to differentiate between standard and deviant sounds (difference wave).

The N250 ERP variables were analysed in our previous publications of JLD ERP findings for children aged 6.5 and 9 years (Hämäläinen et al.,

2013; Lohvansuu et al., 2014). At both time-points, a passive auditory oddball task with naturalistic speech sounds (/ata/ and /ata:/) was administered while children watched a silent movie or played a simple computer game. The speech stimuli presented had no meaning in the Finnish language, although it included a difference in phonemic length, an important semantically distinguishing feature in the Finnish language (Leppänen et al., 2002; Pennala et al., 2010; Richardson et al., 2003). In addition, at the age of 6.5, data collection included complex non-speech tones matched for the frequencies and amplitudes of the harmonics of the speech stimuli. The percentage of presentation for standard and deviant stimuli were kept constant at 80 % and 10 %, respectively, across all paradigms administered; the additional 10 % was dedicated to a second deviant stimulus with a smaller difference to the standard stimulus not examined in the current investigation.

Furthermore, both EEG data collections were performed at the University of Jyväskylä. At 6.5 years, EEG was recorded with 16 Ag/AgCl electrodes at a 200 Hz sampling rate from 88 children. However, due to data quality and other exclusion criteria, only datasets from 64 children were available for inclusion in the current study (more information on the preprocessing of EEG and resulting ERP data can be found in an earlier publication (Hämäläinen et al., 2013)). At 9 years, EEG was recorded with 128 Ag/AgCl electrodes at a 500 Hz sampling rate from a total of 143 children. Due to data quality and other exclusion criteria, only datasets from 138 children were available for modelling purposes (more information on the age 9 EEG and ERP processing is given in an earlier publication (Lohvansuu et al., 2014)). For the proposed models, ERP at age 9 represented the peak amplitudes evoked by the deviant sound (/ata/) in the N250 time window at channel location F3. The model variable ERP at ages 6.5 represented the responses to the deviant

sound (/at:a/) at the N250 time window at the F3 and F4 channel locations, which were extracted by temporal principal component analysis (tPCA). The Cronbach's alpha of the tPC values at channel locations F3 and F4 in the speech stimuli condition was 0.90, while that of the tPC at channel locations F3 and F4 in the non-speech stimuli condition was 0.91. The channel locations F3 and F4 were selected due to the auditory brain response having largest amplitudes at these sites. To facilitate the interpretation of the model, the signs of the brain components with negative polarity were reversed. Thus, larger negative amplitudes were shown as higher numerical values in the statistical models, as the amplitude of the N250 ERP deflection/waveform typically has a negative polarity (having negative numerical values).

2.7. Statistical analysis

To investigate predictive longitudinal associations between children's print exposure (independent and shared with parents) and brain responses and their reading skills (fluency and accuracy), longitudinal path models were constructed using MPlus 7.4 (Muthén & Muthén, 2012). The skewness of the data was addressed using the maximum likelihood estimator with robust standard errors (MLR) (Maydeu-Olivares, 2017; Savalei, 2010). Full information maximum likelihood estimation (FIML) was used to account for missing data. The goodness-of-fit of all models was evaluated with a chi-square test (χ^2 , threshold > 0.05), comparative fit index (CFI, threshold > 0.95), root mean square error of approximation (RMSEA, threshold < 0.06) and standardised root mean square residual (SRMR, threshold < 0.08) (Hu & Bentler, 1999). Modification indices provided in Mplus were examined to optimise model fit. Our analysis strategy consisted of two main steps. In the first step, we constructed a model for each of the speech and non-speech ERP experiments (hypothesised model in Fig. 1). The outcome variables of reading skill, brain response and print exposure at third grade were included in one model to map their associations. To account for the possibility of the models being different between the FR and NFR group children, multiple group comparisons were performed with the Satorra-Bentler corrected χ^2 difference test. The result of this analysis indicated a non-significant difference between the FR and NFR models. Thus, the grouping did not moderate the associations between the model variables and thus the models for the whole sample were reported. In the second step, mediation analysis was performed with JASP 0.17.1 (JASP Team, 2023) based on the lavaan software package (Rosseel, 2012). This was done to further examine the indirect effects that flowed through ERP speech at age 6.5 on early print exposures' relation to Grade 3-level reading skills, unaffected by the other variables entered in the hypothesised model. This analysis was performed to further zoom in to the core relationships among these three variables. The mediation analysis led to four mediation analysis outputs arising from the two types of print exposure (parental/independent) and two types of later reading outcomes (fluency/accuracy). A maximum likelihood estimator was utilized, and missing values were handled with FIML. CI values were computed using the bias-corrected percentile method (Biesanz et al., 2010), including bootstrapping with 1000 replications.

3. Results

3.1. Descriptives and model fit

Table 1 reports the descriptive statistics and correlation coefficients for all the study variables in the hypothesised models. The variables approximated normal distribution except for the reading accuracy variable which was expectedly skewed; most children were close to 100 % accuracy level in Grade 3. Correlation coefficients suggested significant associations within print exposure and ERP domains, but also with across domains and with reading skills. No significant correlations were

found with parental education.

For the speech model the hypothesised model fitted the data well. However, to achieve an acceptable fit with the hypothesised non-speech model, the modification indices were consulted, resulting in an additional path added from parent-child print exposure at ages 4–5 to independent print exposure at ages 9–10. To ensure comparability between the models, the suggested path was added to both models (Fig. 3). The estimated path models yielded a good fit to the data (Supplementary material Table 1). Fig. 3 provides values of the significant standardised path estimates for the speech and non-speech models. All standardised estimates and residual correlations of the models are presented in Supplementary material Tables 2.1 and 2.2.

3.2. ERP speech model

The speech model (Fig. 3a) yielded several interesting findings. First, children's independent print exposure at ages 4–5 showed a significant negative association with ERP responses to speech at age 6–7. In particular, children who spent more time with print materials independently at ages 4–5 had smaller N250 amplitudes to speech stimuli at age 6.5. Second, the ERP responses to speech at age 6–7 were negatively associated with children's independent print exposure at 9 years, indicating that larger N250 responses to speech stimuli were related to less change in time spent with print materials independently. Correlation coefficient (-0.29^*) suggested that in addition, the larger N250 responses to speech stimuli at age 6–7 were related with less independent print exposure at age 9. Third, early (age 4–5) print exposure with parents did not predict children's subsequent brain activation to speech stimuli. However, the timepoint around the onset of formal education in reading (ages 6–7) revealed a significant positive association between parent-child print exposure and the change in ERP towards age 9. Specifically, children who had more parent-child shared reading interactions around the onset of formal reading instruction (ages 6–7), showed a larger change in ERP from age 6.5 to age 9. This indicated that the more time the child spent reading with parents during the transition to formal literacy training, the more negative and larger the N250 responses were compared to the levels they had at age 6.5. Fourth, a higher family risk for dyslexia was directly associated with a more negative amplitude of the brain responses to speech at age 6.5. Fifth, a significant association was found between children's independent print exposure at ages 6–7 and their subsequent reading fluency performance. This meant that the more time spent by children engaged with print independently, the greater their reading fluency skills in Grade 3. Finally, family risk for dyslexia directly and negatively predicted children's later reading outcomes and age 6–7 ERP measure. The model did not reveal any other significant associations from family risk. Parental education was not associated with any of the variables in the model.

3.3. ERP Non-Speech model

The ERP non-speech model at age 6.5 (Fig. 3b) did not differ from the ERP speech model (Fig. 3a) in its main points. However, there were three divergent findings. First, children's independent print exposure at the earliest timepoint (age 4–5) was not associated with the N250 responses to non-speech tones. There was a negative association between ERP non-speech at age 6.5 and children's reading accuracy skills at age 9. Finally, the relationship between family risk and ERP to non-speech stimuli at age 6.5 was non-significant.

3.4. Mediation analysis

In the final stage of the analysis, we conducted mediation modelling to further inspect the core relationships of print exposure at ages 4–5, brain activation to speech sounds at age 6.5 and the later reading skill at

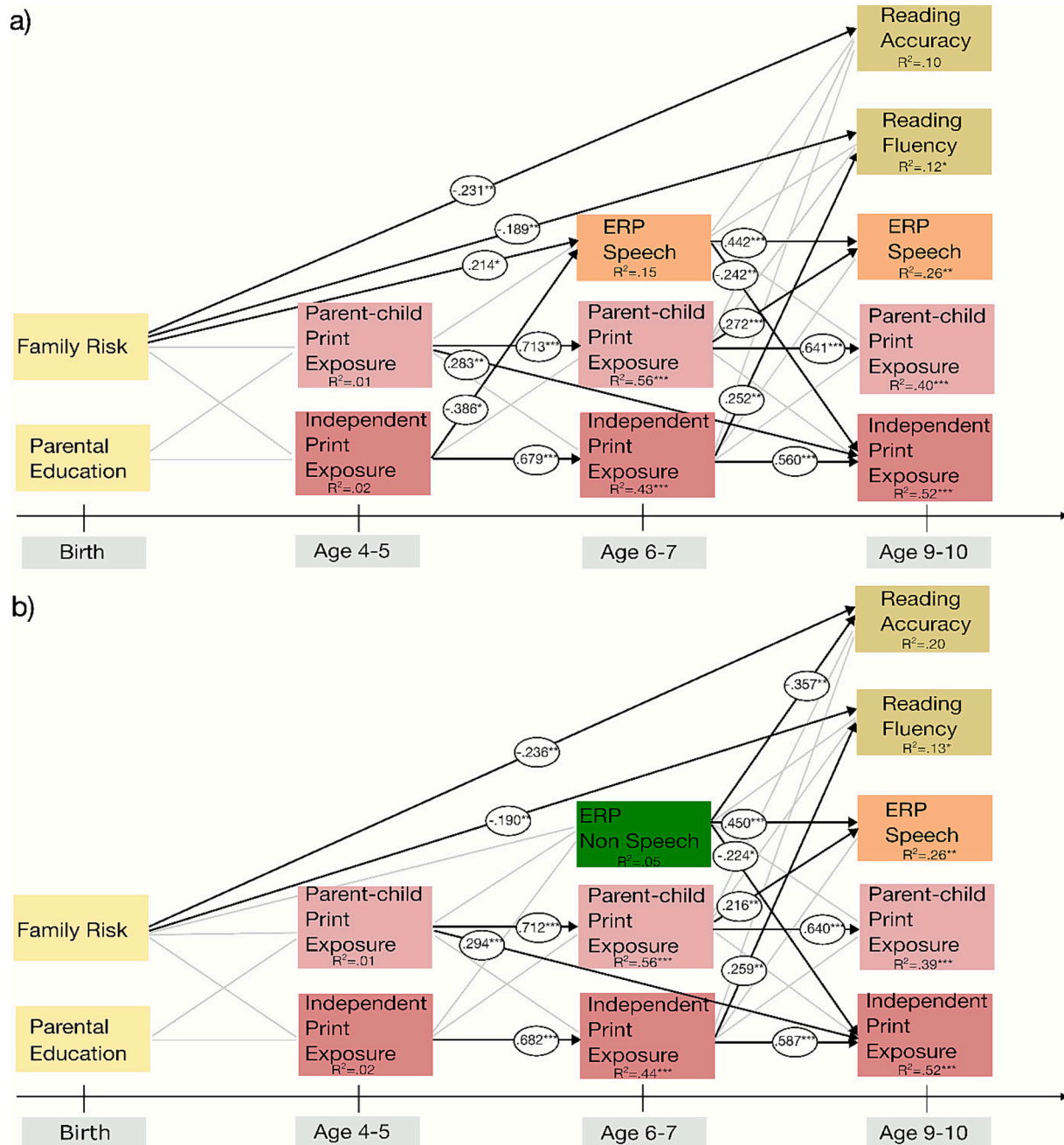


Fig. 3. Model outcomes for (a) speech and (b) non-speech model with significant paths only. A higher value in family risk indicates a risk due to parental dyslexia. To facilitate better model understanding, ERP responses are reversed in amplitude, to reflect a higher amplitude in a rising score. Significance levels are represented by *** $< .001$, ** $< .01$, and * $< .05$. Note that the path from Parent-child Print Exposure 4–5 to Independent Print Exposure 9–10 was added to the hypothesised model to improve the goodness-of-fit for the non-speech model. To enable comparability, the same path was added to the speech model.

Grade 3 away from the other variables entered in the hypothesised model. When examining the mediation outputs (Supplementary material Fig. 1), it can be noted that the direct predictive effects of children's print exposure to Grade 3-level reading accuracy and fluency were non-significant (i.e. independent print exposure to reading fluency: $\beta = -0.039$ (0.124) $p = 0.754$ and parent-child print exposure to reading fluency: $\beta = -0.013$ (0.132) $p = 0.918$). The indirect path via the brain responses to speech was also non-significant despite the significant path coefficients between independent print exposure at ages 4–5 and ERP to speech sounds at age 6.5 (Figs. 3a and b, respectively), as well as the significant path of ERP speech at age 6.5 to reading skill at Grade 3. Thus, despite the individual associations' significance presented in the

longitudinal path model, the mediating role of ERP responses for print exposure and reading fluency lacks support.

4. Discussion

The objective of this study was to longitudinally examine the relationships between print exposure and brain activation pattern previously associated with reading skill. The novel aspect of our models lies in combining EEG data, family risk for dyslexia, questionnaire data on print exposure and reading skills assessments of children, enabling us to display longitudinal associations across the ages 4–10 years. The results revealed that an interplay between print exposure and brain response is

present, although the causality remains unresolved in the presented modelling approach.

4.1. Parent-child print exposure and ERP

In the current study, the association between parent-child shared reading activities and ERP components representing the auditory speech processing is present at the change between age 6–7 to age 9–10, aligning with a recent study where 8 year old children showed a positive relation between frequency of HLE activities and neural activation representing word adaptation in brain regions associated with semantic processing (Girard et al., 2021). Especially in these later time periods of development it's critical to examine the directions of effects in future studies to understand which aspects of the cognitive enrichment of the environment are in fact displayed as a response to the child's development. Meaning that children have so called evocative effects on their environments, which has been found for poor and higher skilled learners for HLE (Salminen et al., 2021; Silinskas et al., 2012, 2013). As shown in our analysis, part of the difference in children's brain responses over time captured in our model was associated with parent-child shared reading at the beginning of formal literacy schooling (ages 6–7). At this timepoint shared reading activities might reflect the first indications that parents react to children's difficulties by providing reading activities at home.

Parental involvement in reading exposure has shown to enhance the processing of print (Zivan & Horowitz-Kraus, 2020) and neural activation of the child during shared reading (Ohgi et al., 2010). In our present study parental print exposure links to the change of independent print exposure over time but lacks the association to ERP response at the first exposure-brain time point. This suggests a discrepancy between our current model, and examinations presenting developmentally earlier associations between neural activation in semantic processing areas and print exposure through shared reading activities. Namely, in 3–5 year olds reading exposure was related to the neural activation (fMRI) measured in brain regions related to semantic language processing (Hutton et al., 2015). In even younger children, about 12 month old infants, extensive shared reading experiences were shown to have an effect on the infant's predictive brain response (fNIRS), which in turn facilitated expressive vocabulary acquisition (Wang et al., 2022). Due to methodological difference in the assessment of neural activation across studies, it remains speculative if the task specificity or the signal modality of the neural measure caused the discrepancy.

4.2. Independent print exposure and ERP

Previous findings have related the concept of reading for pleasure—as a form of independent print exposure in childhood—with overall higher cognitive performance and the adolescent brain structures (Sun et al., 2024). In alignment with this, we report children's independent print exposure to be reciprocally associated with brain activation to speech (even after including familial risk for dyslexia and parental education as controls), suggesting a print exposure – brain association. The ERP response to speech and non-speech stimuli was associated to the change of independent print exposure over time, while a positive relationship was detected from independent print exposure to reading fluency skill at third Grade. Thus, children's independent exposure to print material appears to be elevated when children's reading skill increased and co-aligned with indicators present in the ERP response at 6.5 Years reflecting skill development. Further studies will need to clarify how reading skill related cognitive development relates to this environmental exposure – brain association. For now, the association between independent print exposure and brain response seems to co-occur with reading skill development but the precise interaction is not yet mapped in the here presented modelling approach. The co-occurrence may reflect changes in neural activation caused by skill development, which highlights the need for future models to examine

the skill development to neural activation in a higher frequency to investigate the unique contribution of independent print exposure apart from the child's skill development. Yet again it also has to be evaluated what neural mechanism independent print exposure taps into alongside skill development, as eye tracking research in preschool children suggests a relatively low attending to print compared to pictures when not directed by an adult (Justice et al., 2005). Variability in children's early skill levels may play an important role in independent print exposures mechanism: Evidence for language skill-dependent differences in perceptions of print or picture material has been found in 4 year old children, suggesting that those with higher emerging skills have better opportunities to increase and consolidate their alphabetic knowledge when looking at print materials independently (Evans et al., 2009).

4.3. Divergent findings print exposure types to ERP interplay

The difference between the non-significant association of early print exposure with parents (shared reading) with the 6.5 year brain response, contrary to the significant relationship between a child's independent print exposure and brain response to speech stimuli, raises questions about the different mechanisms at play between shared reading and independent exposure on the developing reader's brain. An earlier study utilizing simultaneous EEG recordings of mothers and children has provided evidence for the neural synchrony of mothers' left hemispheric region of language production to their children's right hemispheric region during shared reading activities (Zivan et al., 2022). Suggestions for the effect of synchronisation of children and their parents during shared reading have been proposed, potentially leading to new ways of classifying the interaction over time (Horowitz-Kraus & Gashri, 2023). Yet, in our presented model, the association fails to show, leaving the transition from evidence seen during shared reading activity to long term associations as a question to be resolve in future investigations. Similarly, independent print exposure must be further evaluated with respect to the process that takes place during exposure and the child's drive leading to independent print exposure. For example, eye tracking combined with neuroimaging could offer opportunities to understand the differences in perception of print materials between shared reading activities and independent print exposure. Our here used method for assessing neural specialization may lack sensitivity for capturing the complex and gradual changes, demanding a more multimodal data collection approach for the developmental trajectories, alongside more nuanced investigation of print exposure.

4.4. Divergent findings print exposure to speech and non-speech model

The divergent associations in the speech and non-speech models between the brain responses and children's independent print exposure at the pre-school ages (4–5 to 6–7) indicate that independent print exposure is more associated with speech processing than general auditory processing at the neural level. However, after school entry, both ERP responses to speech and to non-speech stimuli were associated with independent print exposure, which in turn was correlated to children's reading fluency at the same age. This demonstrates shared effects between more general auditory processing and speech processing in their explanatory value for the change in independent engagement with print material over time. Taken together, it appears that the earlier association of independent print exposure to ERP was speech focused whereas later its shared between speech and general auditory processing. Speech is closely linked to phonological processing, which is known to be deficient in dyslexia, yet the relevance of phonology is known to decrease with age in transparent languages, potentially leading to the later increase in importance of general auditory processing.

Despite the highlighted differences in the environment-brain association, the associations between speech and non-speech brain responses and children's reading skills paralleled some of the findings of previous studies from the same sample, in which brain responses at ages 6.5 and

9 years were associated with reading accuracy in the family risk group (Hämäläinen et al., 2013; Lohvansuu et al., 2014). In our study, earlier ERP responses to non-speech stimuli predicted later reading accuracy, in line with previous findings, as this path remained present, despite entering print exposure measures into the model. Interestingly, the direct path of family risk to later reading skills did not leave significant additional variance in reading accuracy to be explained by speech ERP response (Fig. 3a), contrary to the non-speech ERP response accounting for additional significant variance (Fig. 3b). In other words, ERP responses to speech sounds share more explanatory power with family risk for dyslexia, indicating that the more general auditory processing captures an additional process linked to reading accuracy that goes beyond the processing of speech. The relationship between family risk and brain response to speech strengthens the findings seen in infancy, which relate to the child's family background and its impact on the brain activation patterns recorded with EEG at birth and at 6 months of age (e.g., Lepänen et al., 1997; Lohvansuu et al., 2018).

4.5. Family risk for dyslexia

Yet, family risk for dyslexia, as a representation of parental skills, directly influenced children's later reading performance, not through print exposure or brain response. This finding supports the intergenerational aspect of reading impairment, as proposed in the intergenerational multiple deficit model of dyslexia (van Bergen et al., 2014) and reported in multiple samples (Snowling & Melby-Lervåg, 2016). It is noteworthy that in the current study the status of family risk for dyslexia has been given after more in depth examination of parental skills opposed to being based on questionnaire data only (reading related skill evaluation, see 2.2). Nevertheless, as a continuous measure, parental reading skills might yield additional information compared to the dichotomous risk factor used in this study. In fact, maternal reading fluency has been shown to be positively correlated with children's higher functional brain connectivity of the reading network and other language-related brain regions (Horowitz-Kraus et al., 2018). This discussion is further fueled by findings indicating that parental language skills, rather than family risk, cause variations in the quality of shared reading (Hamilton et al., 2021). In the presented model, family risk was taken into consideration as a means of expressing the genetic source of individual differences, thus, a complete fulfillment of a genetic nurturing design is not given (Hart et al., 2021).

4.6. Conclusion and future directions

In conclusion, the current work encourages future research to combine the niches of research domains present in developmental research (environmental factors, brain activation/responses, genetic variability and skill outcomes). The presented model investigated reading as a continuous outcome variable, along with the brain-environment interaction based on measures of print exposure and the widely observed speech processing deficits in dyslexia (Schulte-Körne & Bruder, 2010). The proposed models displayed a significant reciprocal interplay between print exposure and brain response over time, presenting evidence for the print exposure – brain association. It is limited by its relatively small sample size at some data points, the distal measure of print exposure and its task specific focus for brain activation. The latter two display a discrepancy in the concepts utilized for the model due to their variation in specificity, which should be aligned in future data collections. Despite these shortcomings, the study provides novel insights into the explanatory value of environmental factors, such as home literacy practice (i.e. shared reading) and children's independent print exposure, towards specialised auditory processing, as captured by the ERP response to speech stimuli. It would be beneficial for the field, to further evaluate the interactions between different forms of print exposure and reading outcomes with less specific brain activation patterns (i.e. resting state measures), as well as a variety of task-related

brain activation patterns (i.e. speech processing, print processing). To this end, pioneering works of simultaneous EEG recordings of shared reading or story listening have been completed (Farah et al., 2019; Zivan et al., 2022) but further evidence accumulation is necessary. In the future, longitudinal studies using such techniques could help further understand the long-term effects of home literacy environment practices on children's brain responses related to reading. The ERP presented a unique factor in the model of the present study, emphasising an avenue for future research that should reach beyond the associations presented and evaluate the question of causality. Related to this, research on the effects of environmental factors on reading skills development is suggested to incorporate individual brain activation patterns and vice versa neuroscientists are suggested to incorporate environmental factors in their investigations, closing the gap between fields.

CRedit authorship contribution statement

Ariane Tretow: Writing – original draft, Visualization, Formal analysis, Conceptualization. **Daria Khanolainen:** Writing – review & editing, Formal analysis, Conceptualization. **Jarmo Hämäläinen:** Writing – review & editing, Supervision. **Minna Torppa:** Writing – review & editing, Supervision, Methodology. **Paavo H.T. Leppänen:** Writing – review & editing, Supervision, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data and code availability

The minimum data to support the findings of the model are available on request from the corresponding author (A.T.). The data are not publicly available due to data sharing restrictions imposed on the ongoing longitudinal investigation. However, the model code used for the analysis is available in the [supplementary material](#).

Contributions

A.T. and D.K. conceptualised the research proposal and compiled the data from the existing JLD database. D.K. and A.T. performed the formal analysis, with supervision from the other authors (M.T., P.H.T.L., and J. H.). A.T. performed the main paper writing, compiling of [supplementary information](#), and data visualization. All authors discussed the results and

implications and provided commentaries on the manuscript at all stages. All authors agreed to the submission of the article. Members of the JLD research team (former project leaders Lyytinen, H. & Ahonen, T.; current principal investigators P.H.T.L., M.T., and J.H.; and data manager Eklund, K., along with senior researchers, post docs, doctoral researchers, and research assistants involved) have all contributed to the overall research design and JLD data collection over time.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.bandl.2025.105671>.

Data availability

The data are not publicly available due to data sharing restrictions imposed on the ongoing longitudinal investigation. Stated is given in Manuscript. Minimum supporting data can be requested.

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